
MERN Deployment Guide

CONTENTS

Catalog

01

Introduction

02

MERN Architecture
Review

03

Deployment
Handoff & Security

04

MLOps and
Conclusion



01

Introduction

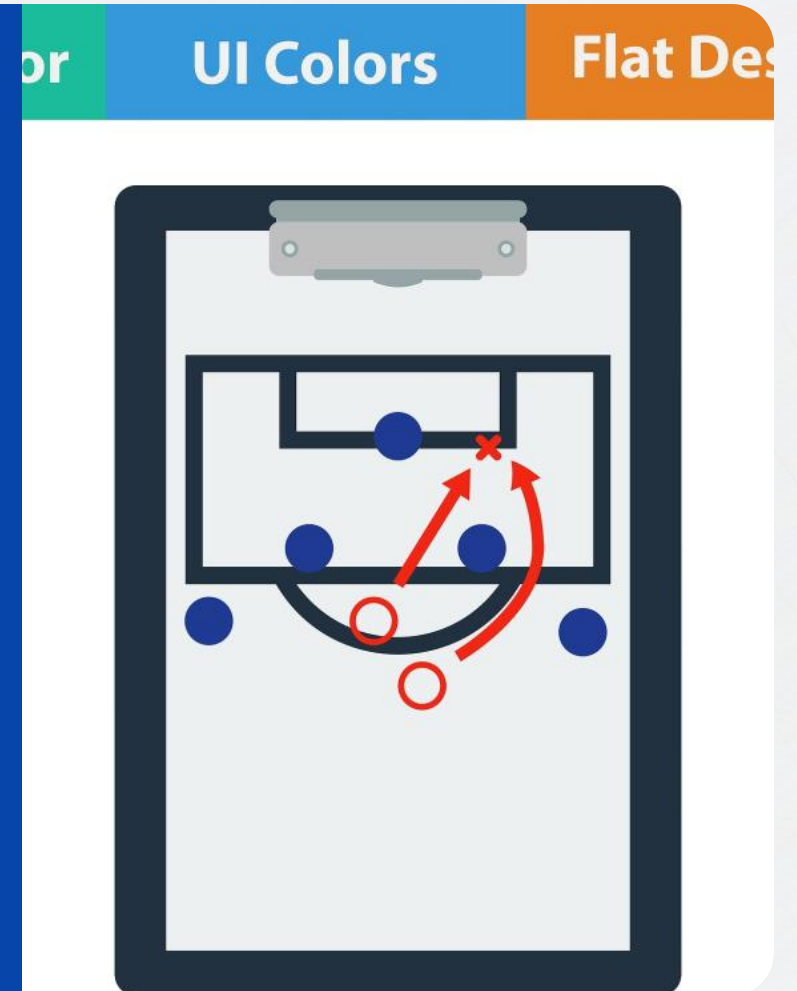
Goal: MERN to live app

Moving prototype to live app

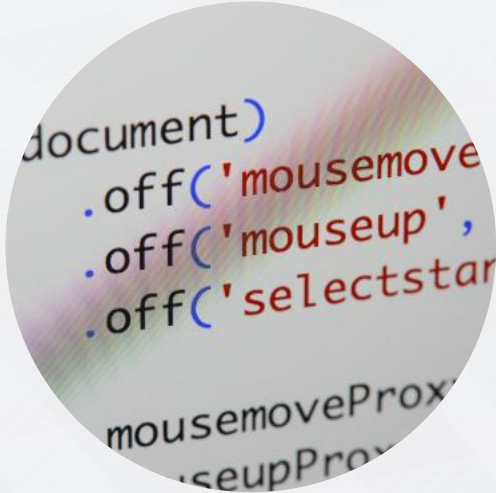
The goal is to transition the MERN prototype into a fully operational, scalable application.

Scalability focus

Emphasizes the importance of making the application live and ready to handle increased traffic and usage.



Defining MERN stack



Unified JavaScript stack

MERN is defined as a stack consisting of MongoDB, Express, React, and Node.js, all using JavaScript.



Comprehensive technology suite

The stack represents a complete set of technologies for building full-stack web applications.

Strategic value for saascv



**Leverage Employees
Potential**

Deployment as critical as training

Deployment strategy is as important as model training for ensuring uptime and security.

Uptime and security emphasis

The strategic value of deploying the MERN stack is highlighted by its impact on the reliability and security of saascv products.

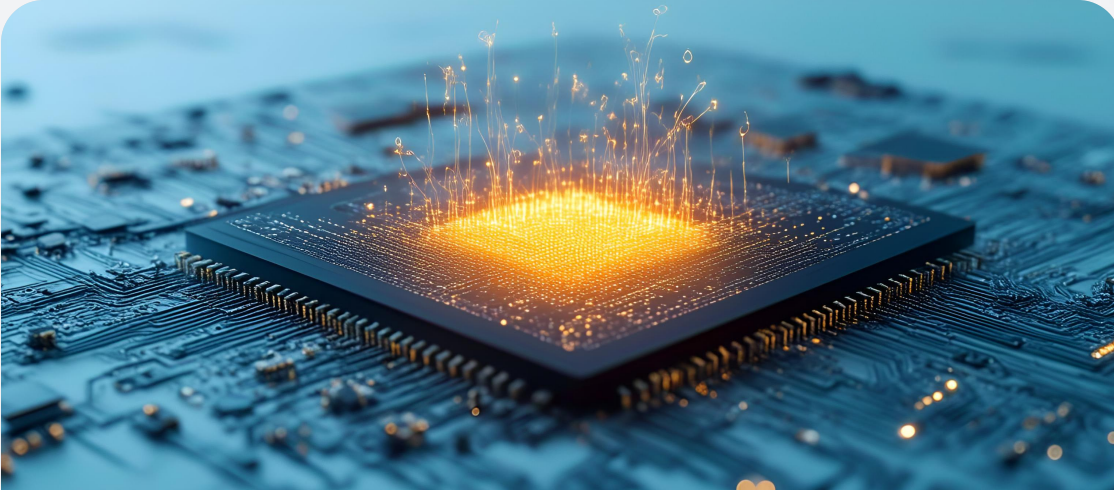
EDITABLE STROKE



02

MERN Architecture Review

Front-end production build

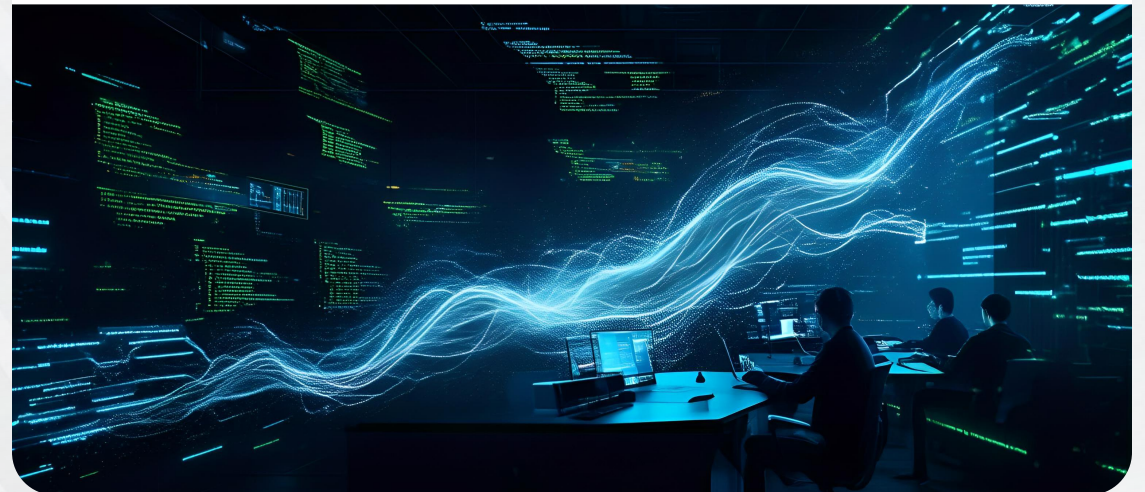


Minified, optimized bundle

Creating a client-side bundle that is minified and optimized for production

Critical for performance

Ensures faster load times and improved user experience in the live application



Backend bridge (Express)



Handling REST API requests

Configuring Express to manage data requests through a RESTful API

Serving static React files

Express also serves the optimized React files for the initial app load

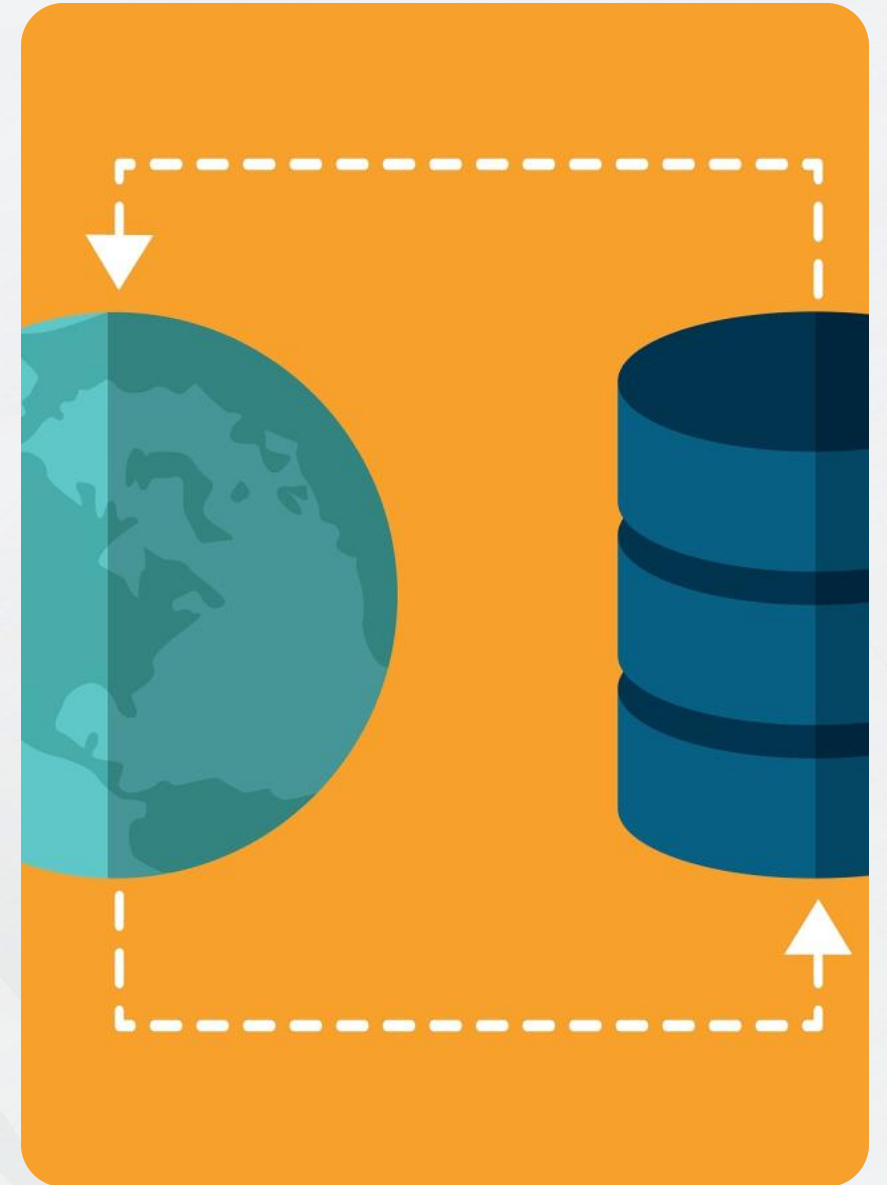
Database configuration

Connecting to MongoDB Atlas

Setting up the Express server to interact with a remote MongoDB Atlas instance

Cloud-hosted database

Utilizing a cloud-hosted solution for scalability and reliability



Environment variables security



Managing secrets in production

Implementing security standards for handling sensitive information like API keys



Baseline security practices

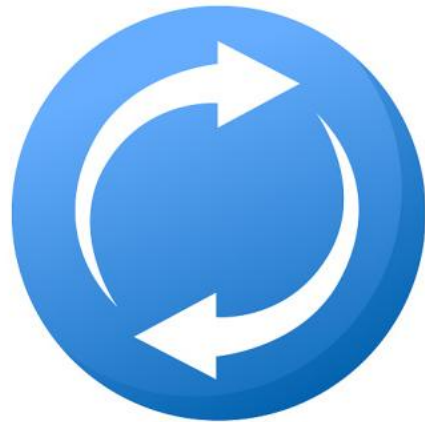
Ensuring environment variables are secured to protect the application's integrity



03

Deployment Handoff & Security

Continuous deployment (CI/CD)



Update...



Automated update systems

Setting up CI/CD pipelines for smooth, frequent updates via GitHub/PaaS.

Frequent updates

Automating deployments to ensure the application stays current and bug-free.

Choosing a host (PaaS)



Simplify server management

Using platforms like Render or Heroku to reduce complexity of server tasks.



Automatic restarts

PaaS providers offer automatic restarts to maintain uptime and reliability.

Securing the API



CORS Configuration

Explicitly allowing production URLs to communicate with the backend for security.



Helmet Middleware

Setting HTTP headers for baseline security against XSS and HSTS attacks.

Scaling fundamentals



Horizontal scaling

Planning for increased traffic by adding more servers to the pool.



PM2 for scaling

Using PM2 for process management to handle increased load efficiently.



04

MLOps and Conclusion



Monitoring & logging

Tools for performance tracking

Implementing tools like Morgan to monitor app performance in production

Error state logging

Utilizing tools to log errors for real-time tracking and historical analysis

Error handling

Graceful failure middleware

Using middleware for controlled error messages and system stability

Dedicated error handling

Ensuring errors are handled with precision to maintain user experience

Conclusion: robust, secure product



Mastery of MERN deployment

Ensuring a robust and secure product through comprehensive deployment mastery

Always-on product delivery

Delivering a reliable, always-on product critical for saascv success

THE END

Thank You